Critical Analysis:
Complexity-Based Prompting for Multi-Step Reasoning

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1 Methodological Strengths

1.1 Clear problem formulation and task focus

The paper provides a focused formulation of the problem of multi-step reasoning with large language models under in-context learning. The study explicitly targets chain-of-thought prompting for math word problems and other symbolic reasoning tasks, and defines a concrete question: which examples and reasoning chains should be selected as prompts and outputs in order to maximise answer accuracy without model fine-tuning. The separation between *prompt selection* in the input space and *consistency selection* in the output space gives the work a coherent methodological structure.

1.2 Simple and interpretable complexity criterion

A key methodological strength is the use of a very simple and interpretable notion of reasoning complexity. Complexity is operationalised as the number of reasoning steps in a chain, where each step is a line separated by a newline. This choice has several advantages:

- It directly aligns with the notion of multi-step reasoning: more steps correspond to deeper or more decomposed reasoning.
- It can be computed automatically once chains are available, without additional models.
- It builds on prior work in reading comprehension that also uses step counts as a proxy for reasoning depth.

By grounding all selection mechanisms in this single scalar quantity, the analysis becomes transparent and reproducible.

1.3 Unified treatment of input and output selection

The methodology extends the complexity-based criterion consistently to both input and output:

- **Complexity-based prompting** selects few-shot examples with the largest number of reasoning steps as in-context demonstrations.
- **Complexity-based consistency** selects, during decoding, the most complex output chains among multiple samples and performs majority voting only on their answers.

This symmetry is conceptually elegant. It shows that the same inductive bias, namely a preference for more elaborate reasoning, can be imposed both when instructing the model and when aggregating its predictions. The design also cleanly generalises the self-consistency framework by viewing it as a special case where all sampled chains share equal weight.

1.4 Careful control of confounders

The empirical design explicitly addresses several potential confounders that might otherwise explain performance improvements:

- **Total number of reasoning steps**: The study constructs prompts that have the same total number of steps but different per-example step counts (such as 24 examples with 3 steps versus 8 examples with 9 steps) and shows that higher per-example step counts yield better accuracy.
- **Prompt length**: Prompts that maximise total character length do not systematically outperform those that maximise reasoning steps per example, which indicates that raw length is not the main driver of gains.
- **Per-step length**: Prompts with extremely short steps underperform those with moderate per-step length, even when the total number of steps is controlled, which supports the view that both step count and per-step informativeness matter.

These controls strengthen the causal interpretation that reasoning step count per example is the dominant factor behind the observed improvements.

1.5 Systematic ablations and robustness checks

The experimental section includes a series of ablations that isolate the contribution of the proposed methods from other prompt-engineering tricks. The breakdown on GSM8K shows that:

- adding the phrase "Let us think step by step" and minor prompt formatting changes provide a first increment in accuracy;
- replacing handcrafted prompts with complex prompts yields an additional and substantial gain;
- restricting majority voting to complex chains produces further improvements on top of self-consistency.

In addition, robustness is tested along several axes:

- **Prompt distribution**: prompts drawn from clean training data, noisy data, and a different dataset all exhibit the same monotonic trend of higher accuracy with more complex examples.
- **Step formatting**: variants that change step delimiters (newline, period, semicolon, explicit "step i") still preserve the advantage of complex prompts despite shifts in absolute performance.
- **Output selection**: voting over most complex chains consistently outperforms voting over all chains, whereas voting over the simplest chains underperforms.

Such robustness checks contribute significantly to the methodological credibility of the findings.

1.6 Breadth of benchmarks and baselines

The study evaluates the proposed approach on three math benchmarks and three non-math reasoning tasks. This includes clean datasets, datasets with noisy labels, and challenging tasks from BigBench Hard. The experiments use multiple model families, including text-davinci-002 and code-davinci-002, and compare against strong baselines such as Minerva, PaLM, and verifier-augmented systems.

Within a single unified framework the paper reproduces prior chain-of-thought and self-consistency baselines, integrates complexity-based prompting, and contrasts it with alternative selection strategies such as centroid-based and retrieval-based selection. This breadth allows a nuanced assessment of where complexity-based prompting is most effective.

1.7 Annotation-efficiency argument

The methodology explicitly considers annotation cost. Complexity-based prompting requires chain-of-thought annotations only for a few complex examples per dataset, which is significantly cheaper than annotating reasoning chains for an entire training set in order to support retrieval-based prompt selection. For datasets that lack chains, the use of question length or formula length as a proxy complexity signal further reduces the annotation burden by focusing manual effort on a small subset of typically eight complex instances. This design keeps the approach close to the spirit of few-shot prompting.

2 Key Limitations

2.1 Narrow task and metric scope

Despite the inclusion of several benchmarks, the evaluation focuses almost entirely on short-answer question answering tasks with accuracy as the sole metric. This leaves several important dimensions unexamined:

- there is no assessment of calibration quality or confidence alignment;
- there is no measure of explanation faithfulness, that is, whether longer chains genuinely reflect the internal reasoning process;
- there is no analysis of robustness under adversarially perturbed inputs.

The exclusive reliance on accuracy obscures potential trade-offs between correctness and explanation quality.

2.2 Simplistic definition of reasoning complexity

Complexity is approximated purely by the number of lines in a reasoning chain. This definition ignores semantic content and coherence. A chain can trivially be made more complex under this metric by splitting sentences or inserting verbose but uninformative steps. The analysis partially mitigates this by controlling per-step length, but does not provide a principled method to distinguish genuinely deeper reasoning from artificially inflated verbosity.

For datasets without gold chains, question length or formula length is used as a proxy, which is only loosely correlated with cognitive difficulty. The approach remains agnostic to whether the task requires combinatorial reasoning, algebraic manipulation, or simple lexical matches.

2.3 Limited error analysis and qualitative understanding

The paper contains minimal qualitative analysis of errors or failure modes. For instance, the following aspects are not explored:

- cases where long chains produce confidently wrong answers;
- patterns of spurious reasoning, such as chains that repeat parts of the problem statement without contributing to computation;
- interactions between complexity and specific operator types, such as division or multi-step temporal offsets.

Without such error analysis, it is difficult to understand whether complexity-based selection primarily reduces shortcut exploitation, or whether it introduces new classes of systematic errors.

2.4 Limited treatment of small models

The experiments with smaller models such as text-curie-001 and Flan-T5 11B are restricted to a small table with three numbers and a brief conclusion that complexity-based prompting is an emergent ability. There is no deeper exploration of why small models fail to benefit, for instance:

- whether small models overfit to surface patterns in complex chains;
- whether temperature or sampling parameters are appropriately tuned for these models;
- whether different complexity ranges might be more suitable for smaller capacities.

This leaves an important part of the model spectrum under-analysed.

2.5 Computational cost and efficiency not quantified systematically

The method relies heavily on self-consistency with fifty samples per test instance and on complex prompts that are longer than typical few-shot examples. The paper briefly mentions monetary cost estimates for validation experiments but does not provide systematic measurements of:

- wall-clock inference time under different values of N and K;
- trade-offs between sampling budget and accuracy;
- memory and latency implications in deployment scenarios.

As a result, the practical feasibility of applying complexity-based consistency at scale remains unclear, especially in resource-constrained settings.

2.6 Reproducibility and dependence on proprietary models

The strongest results depend on proprietary models served through APIs and on numbers for larger models such as PaLM and Minerva that are taken from other papers. While this is common in contemporary large language model research, it introduces reproducibility challenges:

- model versions and underlying training data may evolve over time, affecting performance;
- hyperparameters such as sampling temperature and nucleus probability are not exhaustively reported across all experiments;
- access to models such as PaLM and Minerva is restricted, which prevents external researchers from verifying comparative claims.

These factors complicate precise replication and extension of the reported results.

2.7 Lack of statistical significance analysis

The study reports single-point accuracy estimates on each benchmark. Given the use of sampling and the relatively modest test set sizes for some datasets, performance may exhibit non-trivial variance. The absence of confidence intervals or statistical tests makes it difficult to assess whether some of the observed differences, especially smaller gains on saturated benchmarks, are robust beyond measurement noise.

3 Technical Bottlenecks

3.1 Natural language chains as unconstrained intermediate representations

Chain-of-thought prompting uses free-form natural language as the intermediate representation for reasoning. This has advantages in interpretability, but it also introduces technical bottlenecks:

- there is no guarantee that the chain aligns with a consistent underlying computation;
- numerical operations are executed implicitly by the language model rather than by explicit arithmetic modules;
- long chains can drift off-topic or introduce contradictions without being detected.

Complexity-based selection reinforces a preference for longer chains, but does not impose any formal structure or correctness constraints on these intermediate representations.

3.2 Sampling-based exploration of reasoning space

Complexity-based consistency operates entirely through random sampling of multiple chains followed by majority voting. This creates several bottlenecks:

- the space of possible reasoning chains is enormous, but exploration is limited to a small random subset;
- sampling-based search is blind to chain quality beyond step count and final answer agreement;
- there is no mechanism to revise or refine chains in light of detected inconsistencies.

The method therefore trades off computational resources for probabilistic coverage of reasoning paths, without leveraging more structured search or verification procedures.

3.3 Prompt context limitations and information bottlenecks

The fixed number of in-context examples, typically eight, combined with the context length constraints of large models, imposes an informational bottleneck:

- prompts cannot simultaneously cover a wide diversity of problem types and maintain high per-example complexity;
- choosing only the most complex examples risks under-representing simpler but frequent patterns that appear in real data;
- context saturation may occur when complex examples become extremely long, which can reduce the effective attention devoted to the test question.

The study does not explore adaptive strategies that vary the number or complexity of examples based on test instance difficulty.

3.4 Trade-offs inherent in complexity preference

Always preferring more complex chains introduces specific trade-offs:

- longer chains increase the surface area for errors and hallucinations;
- verbosity can obscure the core reasoning steps and make manual inspection more difficult;
- some tasks may be genuinely solvable with short, direct reasoning, in which case forcing long chains may encourage unnecessary elaboration.

The experiments show gains in aggregate accuracy, but do not characterise scenarios where complexity preference degrades performance, beyond the small drop observed in one cross-dataset setting.

3.5 Integration with external tools and verifiers

The paper briefly compares against systems that incorporate verifiers, but the proposed method itself does not integrate any external checking mechanism. This leaves several technical opportunities unaddressed:

- there is no arithmetic checker to validate intermediate numerical steps;
- there is no logical consistency checker for referential or temporal constraints;
- there is no mechanism to detect and penalise self-contradictory chains.

As a result, the pipeline remains purely generative and cannot fully exploit the potential of symbolic or programmatic verification to narrow the reasoning space.

3.6 Dependence on labelled chains and proxy measures

Although the method is relatively annotation-efficient, it still requires some amount of high-quality chain-of-thought supervision or reliable proxy measures such as question length. For domains where problem texts are short but reasoning is deep, question length may not correlate with complexity. In such settings, the current selection mechanism would struggle to locate genuinely informative examples without additional supervision signals.

4 Research Implications

4.1 Insights into emergent reasoning behaviours

The finding that complexity-based prompting brings substantial gains only for large models provides evidence about the nature of emergent reasoning capabilities. The absence of benefits for smaller models suggests that complex in-context examples are useful only once the model has already internalised sufficiently rich latent computation patterns. This contributes to a broader picture in which in-context learning and multi-step reasoning emerge only beyond certain scale thresholds.

4.2 Prompt engineering as a principled design problem

The work highlights that prompt design for reasoning can be guided by more principled criteria than ad hoc trial-and-error. Instead of searching over arbitrary example sets, practitioners can focus on examples with high reasoning complexity as a robust heuristic. The observation that per-example step count, rather than overall prompt length, is the main determinant of performance clarifies how to allocate limited context budget.

4.3 Revisiting the role of explanation length

The positive impact of longer reasoning chains challenges an intuitive concern that long explanations are necessarily worse due to compounded error probabilities. The results suggest that for sufficiently capable models, longer chains can regularise reasoning by enforcing explicit decomposition. This has implications for the design of explanation-based interfaces in applied systems: requiring models to articulate more detailed reasoning may improve accuracy, not just interpretability.

4.4 Benchmark design and evaluation practices

The paper reinforces the importance of benchmarks that stress multi-step reasoning rather than single-hop pattern recognition. It also illustrates how evaluation should consider not only final answers but also properties of intermediate chains. Future benchmark design can use step count, chain diversity, and robustness to chain perturbations as additional axes of assessment.

4.5 Connections to curriculum learning and training data curation

Complexity-based selection in the in-context setting resonates with ideas from curriculum learning and data curation in supervised training. However, the direction is somewhat inverted: instead of starting with simple examples and gradually increasing difficulty, the model is prompted with the most complex examples available. This suggests a research direction where training curricula and prompting curricula are jointly designed to shape the model's internal representation of reasoning tasks.

5 Potential Research Directions

5.1 Richer and more principled complexity measures

A first avenue of work is to design complexity measures that go beyond simple step counts. Possible directions include:

- structural complexity metrics that capture branching, reuse of intermediate variables, or depth of dependency chains within the reasoning;
- information-theoretic measures that quantify how much new information each step contributes relative to the problem statement and previous steps;
- semantic difficulty estimators learned from data, which can score chains based on alignment with symbolic solvers or verifier outcomes.

Such measures could discriminate between genuinely deep reasoning and superficial verbosity while preserving the advantages of the current approach.

5.2 Quality-aware chain selection and verification

Complexity-based consistency can be extended to incorporate explicit quality signals. For instance:

- numerical consistency checks for arithmetic tasks that verify whether the chain's intermediate computations match the final answer;
- symbolic execution of extracted programs or equations for math and temporal reasoning;

- logical consistency checks for referential and commonsense tasks.

Voting could then be performed over chains that satisfy both complexity thresholds and quality constraints, potentially achieving better accuracy and reliability.

5.3 Adaptive and instance-aware prompting

Rather than using a fixed set of complex examples for all test instances, future work can design adaptive selection mechanisms that consider both instance similarity and complexity. Hybrid strategies may:

- retrieve candidate examples based on semantic similarity and then select the most complex subset among them;
- adjust the number and complexity of prompt examples based on estimated difficulty of the test question;
- learn prompt selection policies with reinforcement learning, guided by validation performance.

Such strategies could preserve the benefits of complexity preference while tailoring prompts to individual inputs.

5.4 Cost-aware inference and sampling

Given the high sampling budget in self-consistency, research is needed on cost-aware variants that maintain most of the accuracy gains with fewer samples. Possible approaches include:

- stopping criteria that terminate sampling once answer distributions concentrate among complex chains;
- importance sampling schemes that bias generation towards complex chains early, reducing the need for many short chains;
- two-stage procedures where a smaller model proposes candidate chains and a larger model refines or filters them.

These methods would improve the practicality of complexity-based consistency in online or large-scale settings.

5.5 Extension to other domains and modalities

The complexity-based principle can be tested beyond text-only math and symbolic reasoning. Candidate domains include:

- code generation, where complexity can be measured by program structure or test coverage;
- multimodal reasoning, where chains might integrate textual and visual evidence over multiple steps;
- interactive decision-making tasks, where chains correspond to action plans over time.

For each domain, suitable complexity metrics and verification tools would need to be developed.

5.6 Theoretical analysis of complexity preference

The empirical findings invite theoretical explanations of why complex prompts and outputs are beneficial. Future work can explore:

- formal models of in-context learning where complex examples constrain the posterior over implicit programs more tightly than simple ones;
- analyses of how self-attention layers process long reasoning chains and how this interacts with parameter scale;
- links between complexity preference and Bayesian marginalisation over latent reasoning paths.

A more rigorous theoretical account would deepen understanding of the mechanisms behind emergent multi-step reasoning in large language models.

6 Conclusion

The paper under analysis introduces a conceptually simple yet empirically strong idea: selecting and trusting reasoning chains based on their complexity, measured by step count. Methodological strengths include a clear problem focus on multi-step reasoning, consistent application of the complexity criterion to both prompts and outputs, careful control of confounding factors, and an extensive empirical evaluation across several benchmarks and model families. These choices support the claim that complex examples and chains play a privileged role in eliciting accurate reasoning from large language models.

At the same time, several limitations and bottlenecks remain. Complexity is measured with a coarse proxy that does not distinguish genuine reasoning depth from verbosity, error analysis is limited, computational costs are not systematically quantified, and the approach shows benefits primarily for very large models. The pipeline remains purely generative, without explicit integration of verification or symbolic reasoning modules.

The most promising research directions include developing richer complexity metrics, combining complexity-based selection with explicit verifiers, designing adaptive and cost-aware prompting and decoding schemes, and extending the principle to new domains and modalities. Progress along these lines can transform complexity-based prompting from a heuristic into a broader framework for structuring and constraining the reasoning processes of large language models.